

Audit Service Documentation

Purpose

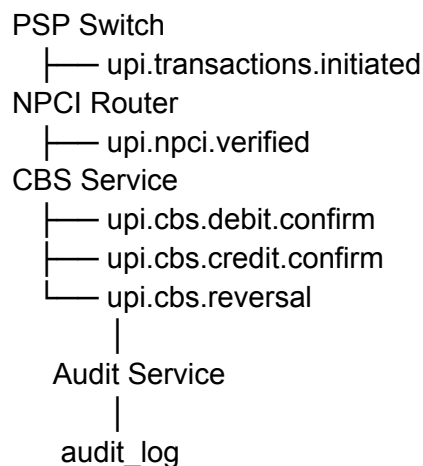
The Audit Service acts as a centralized transaction monitoring and audit trail system. Unlike PSP Ledger, NPCI Ledger, and Bank Ledger which store specific business records, the Audit Service records the complete lifecycle of a transaction across all services.

It provides:

- Transaction traceability
- Event monitoring
- Failure investigation
- Compliance auditing
- Reversal tracking
- Historical transaction logs

The Audit Service consumes events from multiple Kafka topics and stores them in a single audit table.

Architecture



Kafka Topics Consumed

The Audit Service listens to all major transaction events.

Topic	Producer Service	Purpose
upi.transactions.initiated	PSP Switch	Payment initiated
upi.npci.verified	NPCI Router	Transaction verified
upi.cbs.debit.confirm	CBS Service	Debit successful
upi.cbs.credit.confirm	CBS Service	Credit successful
upi.cbs.reversal	NPCI Router / CBS	Reversal initiated

Kafka Message Format

The Audit Service expects messages containing the following fields.

```
{
  "txnId": "TXN1001",
  "source": "CBS",
  "status": "SUCCESS",
  "payer": "alice@sbi",
  "payee": "bob@hdfc",
  "amount": 500,
  "stage": "DEBIT",
  "remarks": "Debit completed successfully"
}
```

Event Examples

PSP Event

Topic:

upi.transactions.initiated

Produced by: PSP Switch

Message:

```
{
  "txnId":"TXN1001",
  "source":"PSP",
  "status":"INITIATED",
  "payer":"alice@sbi",
  "payee":"bob@hdfc",
  "amount":500,
  "stage":"INITIATION",
  "remarks":"Payment initiated"
}
```

NPCI Verification Event

Topic: upi.npci.verified

Produced by: NPCI Router

Message:

```
{
  "txnId":"TXN1001",
  "source":"NPCI",
  "status":"VERIFIED",
  "payer":"alice@sbi",
  "payee":"bob@hdfc",
  "amount":500,
  "stage":"ROUTING",
  "remarks":"Transaction verified"
}
```

Debit Confirmation Event

Topic: upi.cbs.debit.confirm

Produced by: CBS Service

Message:

```
{
  "txnId":"TXN1001",
  "source":"CBS",
  "status":"SUCCESS",
}
```

```
"payer":"alice@sbi",
"payee":"bob@hdfc",
"amount":500,
"stage":"DEBIT",
"remarks":"Amount debited"
}
```

Credit Confirmation Event

Topic: upi.cbs.credit.confirm

Produced by: CBS Service

Message:

```
{
  "txnId":"TXN1001",
  "source":"CBS",
  "status":"SUCCESS",
  "payer":"alice@sbi",
  "payee":"bob@hdfc",
  "amount":500,
  "stage":"CREDIT",
  "remarks":"Amount credited"
}
```

Reversal Event

Topic: upi.cbs.reversal

Produced by: NPCI Router

Message:

```
{
  "txnId":"TXN1001",
  "source":"NPCI",
  "status":"REVERSED",
  "payer":"alice@sbi",
  "payee":"bob@hdfc",
  "amount":500,
  "stage":"REVERSAL",
  "remarks":"Auto reversal triggered"
}
```

PostgreSQL Table

Table Name

audit_log

Create Table Script

```
CREATE TABLE audit_log (  
  id SERIAL PRIMARY KEY,  
  
  txn_id VARCHAR(100),  
  
  source VARCHAR(50),  
  
  status VARCHAR(50),  
  
  payer_vpa VARCHAR(100),  
  
  payee_vpa VARCHAR(100),  
  
  amount DOUBLE PRECISION,  
  
  stage VARCHAR(50),  
  
  remarks VARCHAR(255),  
  
  created_at TIMESTAMP  
);
```

Table Structure

Column	Description
id	Auto-generated Primary Key
txn_id	Unique Transaction ID
source	Event producer (PSP/NPCI/CBS)
status	INITIATED / VERIFIED / SUCCESS / REVERSED
payer_vpa	Sender UPI ID
payee_vpa	Receiver UPI ID

amount	Transaction Amount
stage	INITIATION / ROUTING / DEBIT / CREDIT / REVERSAL
remarks	Additional information
created_at	Event timestamp

Sample Audit Trail

For transaction TXN1001:

txn_id	source	status	stage
TXN1001	PSP	INITIATED	INITIATION
TXN1001	NPCI	VERIFIED	ROUTING
TXN1001	CBS	SUCCESS	DEBIT
TXN1001	CBS	SUCCESS	CREDIT

If credit fails:

txn_id	source	status	stage
TXN1001	PSP	INITIATED	INITIATION
TXN1001	NPCI	VERIFIED	ROUTING
TXN1001	CBS	SUCCESS	DEBIT
TXN1001	NPCI	REVERSED	REVERSAL